The First Pass: A Complete Worked Example Through the Generative-Collapse Spine

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We trace a single measurement—three voltage readings at 99% fidelity—through the complete validation spine of the Universal Measurement Contract Protocol (UMCP). Starting from Axiom-0 ("Collapse is generative; only what returns is real") and a frozen contract, we embed the raw data into a bounded trace vector $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^3$, compute the six kernel invariants $(F, \omega, S, C, \kappa, \mathrm{IC})$, reconcile the integrity ledger, classify the regime via four frozen gates, and derive the three-valued verdict CONFORMANT. Every number is exact, every identity is verified to machine precision, and every step is reproducible from the open-source reference implementation. We present the kernel computation in explicit detail, prove the three structural identities ($F+\omega = 1$, $\mathrm{IC} \leq F$, $\mathrm{IC} = e^{\kappa}$) for the worked example, demonstrate the seam budget closure, and show how the same spine applies identically regardless of the interface used—command line, Python API, or REST endpoint. A heterogeneous counterexample is provided to demonstrate geometric slaughter: the mechanism by which a single degraded channel destroys multiplicative coherence while leaving the arithmetic mean apparently healthy. The paper serves as a self-contained entrance to the GCD formalism, requiring no prior knowledge of the framework.

I. INTRODUCTION

The Generative Collapse Dynamics (GCD) framework and its operational arm, the Universal Measurement Contract Protocol (UMCP) [1, 2], constitute a contract-first validation system built on a single axiom. The framework has been applied across 23 domain closures [3], from particle physics [2] to consciousness coherence, with 20,221 automated tests, 47 lemmas, and 44 structural identities verified to machine precision.

Despite this scope, no single document has previously traced one complete object through every layer of the system: from the axiom through the frozen contract, embedding, kernel computation, ledger reconciliation, regime classification, casepack packaging, and operational execution—all with exact numbers.

This paper provides that trace. The object is deliberately minimal: three identical voltage readings. The simplicity is the point. When the system is understood for the simplest case, the general case follows by replacing the trace vector—the spine does not change.

A. Scope and structure

We present twelve stops, organized into the five stages of the fixed discourse spine:

1. **Contract** (Stops 1–2): The axiom is stated; the contract is frozen.
2. **Canon** (Stops 3–5): Channels are chosen, the trace is bounded, and the kernel is computed.
3. **Closures** (Stop 6): Thresholds are published; the ledger is debited and credited.
4. **Integrity Ledger** (Stops 6–7): The seam residual is checked; the regime is classified.
5. **Stance** (Stops 7–12): The verdict is derived; the result is packaged, executed, rendered, and read back into ordinary language.

Figure 1 maps the twelve stops onto the five spine stages visually.

Stop	What happens	Spine stage
1	Axiom declared	CONTRACT
2	Parameters frozen	
3	Channels chosen	CANON
4	Trace bounded	
5	Kernel computed	
6	Ledger debited / credited	CLOSURES
7	Regime classified	INTEGRITY LEDGER
8	Casepack assembled	STANCE
9	CLI executed	
10	API called	
11	Latin rendered	
12	Ordinary reading	

FIG. 1. The twelve stops mapped onto the five stages of the fixed discourse spine: Contract → Canon → Closures → Integrity Ledger → Stance.

B. Source files

The worked example traces through the following repository paths (relative to the project root):

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Role	Path
Casepack	casepacks/hello_world/
Contract	contracts/UMA.INTSTACK.v1.yaml
Kernel code	src/umcp/kernel_optimized.py
Seam budget	src/umcp/seam_optimized.py
Frozen params	src/umcp/frozen_contract.py
Validator	src/umcp/validator.py
CLI entry	src/umcp/cli.py
This guide	THE_FIRST_PASS.md

Appendix A provides a heterogeneous counterexample. Appendix B establishes the precise relationship between the GCD kernel and classical results.

II. THE AXIOM

Axiom (Axiom-0)**.** Collapse is generative; only what returns is real.

Collapsus generativus est; solum quod redit, reale est.

This is the single foundational principle. It is a constraint on admissible claims: no measurement, computation, or verdict is admitted unless it can demonstrate *return*—meaning the system can re-enter its admissible neighborhood D_θ after drift, perturbation, or delay, under the same evaluation rules that were declared before the data was seen.

Formally, return at time t requires the existence of a prior state $u \in D_\theta(t)$ such that $\|\Psi(t) - \Psi(u)\| \leq \eta$, where η is the frozen return tolerance. When return is indefinitely deferred, $\tau_R = \infty_{\text{rec}}$, and no epistemic credit is granted.

Everything that follows derives from this axiom.

III. THE FROZEN CONTRACT

Before touching data, we declare the rules. The contract UMA.INTSTACK.v1 [4] freezes all parameters prior to computation.

Definition 1 (Frozen parameters)**.** *The frozen parameters are:*

$$\begin{aligned} \varepsilon &= 10^{-8} && \textit{(guard band)}, \\ p &= 3 && \textit{(drift cost exponent)}, \\ \alpha &= 1.0 && \textit{(curvature cost coefficient)}, \\ \text{tol}_{\text{seam}} &= 0.005 && \textit{(seam residual tolerance)}. \end{aligned}$$

These values are not arbitrary. They are *seam-derived*: the unique constants where mathematical identities hold consistently across all 23 tested domains [2, 3]. The exponent $p = 3$ is the unique integer for which ω_{trap} is a Cardano root of $x^3 + x - 1 = 0$. The tolerance $\text{tol}_{\text{seam}} = 0.005$ is the width at which $\text{IC} \leq F$ holds at 100% across every domain. The guard band $\varepsilon = 10^{-8}$ is chosen so that the pole at $\omega = 1$ does not affect any measurement to machine precision.

The contract also declares three algebraic identities that every computation must satisfy:

$$F + \omega = 1 \qquad \text{(duality)}, \tag{1}$$

$$\text{IC} \leq F \qquad \text{(integrity bound)}, \tag{2}$$

$$\text{IC} = e^{\kappa} \qquad \text{(log-integrity)}. \tag{3}$$

And three-valued verdicts: CONFORMANT | NONCONFORMANT | NON-EVALUABLE. Never boolean.

IV. CHANNEL CHOICE AND EMBEDDING

At $t = 0$, a sensor reports three raw values:

$$x_1 = 9.9, \quad x_2 = 9.9, \quad x_3 = 9.9, \tag{4}$$

with declared instrument range $[0, 10]$.

Channel choice is a Tier-2 act: the domain closure maps physical quantities to the unit interval. With linear normalization:

$$c_k = \frac{x_k - x_{\min}}{x_{\max} - x_{\min}} = \frac{x_k}{10}, \tag{5}$$

yielding $c_1 = c_2 = c_3 = 0.99$. Equal weights $w_k = 1/3$ are assigned (no domain-specific reason to privilege any channel).

Definition 2 (Trace and weight vectors)**.** *The trace vector is* $\mathbf{c} = (0.99, 0.99, 0.99) \in [0, 1]^3$. *The weight vector is* $\mathbf{w} = (1/3, 1/3, 1/3) \in \Delta^2$. *Both satisfy the contract constraints:* $c_k \in [\varepsilon, 1 - \varepsilon]$ *and* $\sum_k w_k = 1$.

A. The guard band

The contract specifies face policy pre_clip: before any computation, every channel is clamped to $[\varepsilon, 1 - \varepsilon]$. For $c_k = 0.99$, the clamp is inactive. The guard band exists because the kernel requires $\ln(c_k)$; if $c_k = 0$, the kernel diverges. The ε-clamp is a return guarantee: no channel can fully die, because a dead channel has no path back through collapse.

If a raw value exceeds the declared bounds (e.g., $x_1 = 12.5$ when the range is $[0, 10]$), the embedding produces $c_1 = 1.25$, which clips to $c_1 = 1 - \varepsilon \approx 1.0$, and an out-of-range (OOR) flag is set. The data is preserved; the flag is the audit trail.

For our trace: no OOR flags. The bounded trace is

$$\Psi(0) = (0.99, 0.99, 0.99). \tag{6}$$

V. THE KERNEL

The kernel $K : [0, 1]^n \times \Delta^{n-1} \to \mathbb{R}^6$ maps the trace and weight vectors to six invariants. We compute each explicitly.

Definition 3 (GCD Kernel)**.** *Given* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and* $\mathbf{w} = (w_1, \ldots, w_n)$ *with* $\sum_i w_i = 1$*:*

$$F = \sum_i w_i\, c_i, \tag{7}$$

$$\omega = 1 - F, \tag{8}$$

$$\kappa = \sum_i w_i \ln c_i, \tag{9}$$

$$\mathrm{IC} = e^{\kappa}, \tag{10}$$

$$S = -\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big], \tag{11}$$

$$C = \frac{\mathrm{stddev}(c_i)}{0.5}. \tag{12}$$

A. Fidelity and drift

Fidelitas (F) measures what survives collapse:

$$F = \tfrac{1}{3}(0.99 + 0.99 + 0.99) = 0.99. \tag{13}$$

Derivatio (ω) measures what is lost:

$$\omega = 1 - 0.99 = 0.01. \tag{14}$$

Proposition 4 (Duality verification)**.** $F + \omega = 0.99 + 0.01 = 1.0$ *exactly. The residual is* $|F + \omega - 1| = 0.0$.

This is not approximate. The duality identity (1) holds to 0.0×10^0 residual across all tested traces [3].

B. Log-integrity and integrity

Log-Integritas (κ):

$$\kappa = \tfrac{1}{3}\big[\ln(0.99) + \ln(0.99) + \ln(0.99)\big] = -0.01005034. \tag{15}$$

$\kappa \le 0$ always, since $\ln(c_i) \le 0$ for $c_i \in (0, 1]$.

Integritas Composita (IC):

$$\mathrm{IC} = e^{-0.01005034} = 0.99. \tag{16}$$

The log-integrity relation (3) holds by construction.

Proposition 5 (Integrity bound verification)**.** $\mathrm{IC} = 0.99 \le 0.99 = F$*. For homogeneous traces* ($c_1 = c_2 = \cdots = c_n$)*, equality holds:* $\mathrm{IC} = F$.

The integrity bound (2) is the solvability condition for trace recovery: given (F, IC) with $n = 2$, the individual channels satisfy $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$, which requires $\mathrm{IC} \le F$ for real solutions. This is derived independently from Axiom-0; the classical AM-GM inequality is the degenerate limit when channel semantics, weights, and the guard band are removed.

C. Entropy

The Bernoulli field entropy (not Shannon entropy, which is the degenerate limit when the collapse field is set to $\{0, 1\}$):

$$\begin{aligned} S &= -\tfrac{1}{3}\big[0.99\ln(0.99) + 0.01\ln(0.01)\big] \times 3 \\ &= 0.05600. \end{aligned} \tag{17}$$

This is low: the channels are near-maximal, leaving little uncertainty in the collapse field.

D. Curvature

Curvatura (C) measures channel spread:

$$C = \frac{\mathrm{stddev}(0.99, 0.99, 0.99)}{0.5} = \frac{0.0}{0.5} = 0.0. \tag{18}$$

With identical channels, there is no heterogeneity.

E. The heterogeneity gap

The gap $\Delta = F - \mathrm{IC}$ measures how much the arithmetic mean conceals about channel dispersion:

$$\Delta = 0.99 - 0.99 = 0.0. \tag{19}$$

For homogeneous traces, $\Delta = 0$. This will become nontrivial in Appendix A.

F. Kernel summary

TABLE I. Kernel outputs for $\mathbf{c} = (0.99, 0.99, 0.99)$ with equal weights.

Symbol	Latin	Value	Meaning
F	Fidelitas	0.99	What survives
ω	Derivatio	0.01	What is lost
κ	Log-Integritas	-0.01005	Log coherence
IC	Integritas	0.99	Mult. coherence
S	Entropia	0.05600	Field entropy
C	Curvatura	0.0	Channel spread
Δ	—	0.0	Heterogeneity gap

VI. THE INTEGRITY LEDGER

The integrity ledger tracks debits and credits across the seam [5]:

Definition 6 (Seam budget)**.** *The budget identity is*

$$\Delta\kappa = R \cdot \tau_R - (D_\omega + D_C), \tag{20}$$

where the drift cost is

$$D_\omega = \Gamma(\omega) = \frac{\omega^p}{1-\omega+\varepsilon} = \frac{(0.01)^3}{0.99+10^{-8}} = 1.01\times 10^{-6}, \tag{21}$$

and the curvature cost is

$$D_C = \alpha \cdot C = 1.0 \times 0.0 = 0.0. \tag{22}$$

At $t = 0$, there is no prior history. The system has never departed and never returned, so $\tau_R = \infty_{\text{rec}}$—infinite return delay. Under the contract's censoring rule (*Si* $\tau_R = \infty_{\text{rec}}$, *nulla fides datur*), no credit is earned:

$$R \cdot \tau_R = 0. \tag{23}$$

The ledger:

Entry	Type	Amount
Drift cost D_ω	Debit	1.01×10^{-6}
Curvature cost D_C	Debit	0.0
Return credit	Credit	0.0
Residual $\lvert s\rvert$		1.01×10^{-6}

The seam closes: $|s| = 1.01\times 10^{-6} \le 0.005 = \text{tol}_{\text{seam}}$.

VII. REGIME CLASSIFICATION

The regime is classified by four conjunctive gates applied to the kernel outputs. These gates are frozen in the contract [4]:

Definition 7 (Regime gates)**.**

$$\begin{aligned}
&\textsc{Stable}: && \omega < 0.038 \ \wedge\ F > 0.90 \\
& && \wedge\ S < 0.15 \ \wedge\ C < 0.14, \\
&\textsc{Watch}: && 0.038 \le \omega < 0.30 \\
& && \textit{(or Stable gates not all met)}, \\
&\textsc{Collapse}: && \omega \ge 0.30, \\
&\textsc{Critical}: && \text{IC} < 0.30 \ \textit{(overlay)}.
\end{aligned}$$

For our trace:

Gate	Threshold	Value	Pass?
$\omega < 0.038$	0.038	0.01	✓
$F > 0.90$	0.90	0.99	✓
$S < 0.15$	0.15	0.056	✓
$C < 0.14$	0.14	0.0	✓

All four gates pass. Regime: Stable. Critical overlay: inactive (IC = 0.99 > 0.30). Stance: conformant.

Remark 1. *In Fisher coordinate space ($\theta = \arcsin\sqrt{\omega}$), the* Stable *regime occupies only 12.5% of the total manifold. The remaining 87.5% is* Watch *(24.4%) and* Collapse *(63.1%). Stability is structurally rare.*

VIII. OPERATIONAL EXECUTION

The preceding computation is packaged in a *casepack*—the atomic unit of auditable work [4]. A casepack is a directory containing the raw data, the frozen contract, the closure registry, and the expected outputs (embedded trace, invariants, SS1M receipt with SHA-256 checksums).

A. Command-line validation

```
$ umcp validate casepacks/hello_world \
    --strict
UMCP validation: CONFORMANT
  (repo + casepacks/hello_world),
  errors=0 warnings=0;
  validator=umcp-validator v2.3.1;
  policy strict=true;
  sha256=6fc79df4...
```

The validator performs, in sequence: JSON Schema validation (Draft 2020-12), semantic rule checks (15+ rules), kernel identity checks ($F = 1-\omega$, $\text{IC} \approx e^\kappa$, $\text{IC} \le F$), regime consistency verification, and SHA-256 integrity checking. Exit code: 0 = conformant, 1 = nonconformant.

B. Programmatic API

The same computation is available via the Python API:

```
from umcp import validate
result = validate("casepacks/hello_world")
assert result["run_status"] == "CONFORMANT"
```

Or via the `MeasurementEngine` for raw data:

```
from umcp.measurement_engine \
    import MeasurementEngine
import numpy as np

engine = MeasurementEngine()
raw = np.array([[9.9, 9.9, 9.9]])
result = engine.from_array(
    raw, weights=[1/3, 1/3, 1/3]
)
row = result.invariants[0]
# row.F == 0.99, row.regime == "Stable"
```

C. The cognitive equalizer

Every execution path—CLI, Python API, REST endpoint, C99 core, C++ accelerator—passes through the same spine with the same frozen parameters and produces the same verdict. This is the *Aequator Cognitivus*: the system produces identical output regardless of which cognitive agent operates it. Thresholds are frozen, vocabulary is operationally defined, and verdicts are derived from gates. The structure measures; the agent does not.

IX. LATIN RENDERING AND CROSS-DOMAIN TRANSLATION

The canonical Latin vocabulary carries operational constraints that English does not. Each term is not a translation but a precision instrument whose morphology encodes the operational definition.

The worked example, rendered canonically:

> *Ad tempus* $t = 0$*, tria vestigia mensurantur:* $\mathbf{c} = (0.99, 0.99, 0.99)$*. Fidelitas est* $F = 0.99$*; derivatio est* $\omega = 0.01$*. Complementum perfectum tenet:* $F + \omega = 1$*. Log-integritas est* $\kappa = -0.01005$*; integritas composita est* $\mathrm{IC} = 0.99$*. Limbus integritatis tenet:* $\mathrm{IC} \leq F$*. Entropia campi est* $S = 0.056$*; curvatura est* $C = 0.0$*. Moratio reditus est* $\tau_R = \infty_{\mathrm{rec}}$*— nulla fides datur. Regio est Stabilis. Sutura clauditur. Iudicium:* CONFORMANT.

A. The five-word vocabulary

The canonical prose interface uses five words, each with a ledger role [1]:

Latin	English	Value
Derivatio	Drift	$\omega = 0.01$
Fidelitas	Fidelity	$F = 0.99$
Curvatura	Roughness	$C = 0.0$
Moratio Reditus	Return	$\tau_R = \infty_{\mathrm{rec}}$
Integritas	Integrity	$\mathrm{IC} = 0.99$

B. Rosetta translation

The Rosetta adapter [1] maps the five words across lenses so different fields can read the same computation in their own dialect:

Lens	Drift = 0.01	Fidelity = 0.99
Physics	1% energy loss	99% preserved
Finance	1% drawdown	99% retained
Epistemology	Minimal shift	Warrant preserved

The meanings of the columns remain stable while the dialect changes. *Significatio stabilis manet dum dialectus mutatur.*

X. DEGENERATE LIMITS

The GCD kernel structures relate to classical results as follows: the classical result is what remains when degrees of freedom are removed. The arrow of derivation runs from the axiom to the classical limit, never in reverse.

Proposition 8 (Classical degenerate limits). *(i)* ***Duality → unitarity.*** $F + \omega = 1$ *reduces to* $p + q = 1$ *when the cost function* $\Gamma(\omega)$ *is removed.*

(ii) ***Integrity bound → AM-GM inequality.*** $\mathrm{IC} \leq F$ *reduces to* $\mathrm{GM}(c) \leq \mathrm{AM}(c)$ *when channel semantics, weights, and the guard band* ε *are removed.*

(iii) ***Bernoulli field entropy → Shannon entropy.*** *The entropy* S *of Eq.* (11) *reduces to Shannon entropy when the collapse field is removed (i.e.,* $c_i \in \{0, 1\}$ *only).*

(iv) ***Log-integrity → exponential map.*** $\mathrm{IC} = e^{\kappa}$ *reduces to the standard exponential map when the weighted sum structure is removed.*

The GCD framework does not "use" or "apply" these classical results. It derives the kernel identities independently from Axiom-0; the classical results emerge as structurally impoverished special cases.

XI. DISCUSSION

We have traced three voltage readings—the simplest possible measurement—through the complete GCD/UMCP spine. The trace demonstrates:

1. **Contract precedence.** All parameters are frozen before any data is seen. No threshold, tolerance, or gate is tuned after evidence.

2. **Exact identities.** $F + \omega = 1$ holds to 0.0×10^0 residual; $\mathrm{IC} = e^{\kappa}$ holds to machine precision; $\mathrm{IC} \leq F$ with equality for homogeneous traces.

3. **Three-valued logic.** The verdict is CONFORMANT, NONCONFORMANT, or NON-EVALUABLE—never boolean.

4. **Typed boundaries.** $\tau_R = \infty_{\mathrm{rec}}$ is a first-class semantic value (not an error or a missing datum) that directly determines ledger credit.

5. **Interface invariance.** CLI, Python API, REST API, C99 core, and C++ accelerator all produce identical results from the same spine and frozen parameters.

6. **Structural stability is rare.** STABLE occupies 12.5% of Fisher space; the homogeneous high-fidelity trace sits in the small corner where all four gates simultaneously pass.

The simplicity of the example is a feature: the spine is identical for three voltage readings and for 31 Standard Model particles [2]. Only the Tier-2 channel choice changes; the Tier-1 kernel and Tier-0 protocol are invariant.

A. Reproduce this pass

Every number in this paper can be regenerated from the open-source repository [3]. The following commands suffice:

Run This Now

Clone & install

```
git clone https://github.com/
    calebpruett927/
    GENERATIVE-COLLAPSE-DYNAMICS.git
cd GENERATIVE-COLLAPSE-DYNAMICS
pip install -e ".[all]"
```

Validate the hello_world casepack

```
umcp validate casepacks/hello_world
    --strict
```

Run the orientation (10 sections)

```
python scripts/orientation.py
```

Run the full test suite

```
pytest
```

XII. CONCLUSIONS

We have provided a complete, self-contained trace of a single measurement through the twelve stops of the GCD validation spine. Every number is reproducible by running the open-source reference implementation [3]. The paper serves as the canonical entrance to the framework: a reader who has seen one complete pass can navigate any domain closure, any theorem, and any extension of the system.

The first pass is complete. Now proceed through any door.

Finis primi transitus. Nunc per quamlibet portam procede.

ACKNOWLEDGMENTS

This work was conducted independently without institutional affiliation or external funding. The kernel's mathematical substrate has a long history: Jakob Bernoulli's *Ars Conjectandi* (1713) established the field structure from which the entropy formula descends; R. A. Fisher's geometric treatment of statistical families (1925) provides the coordinate space in which regime partition is measured; Claude Shannon's channel capacity theorem (1948) is recovered here as a degenerate limit when the collapse field is removed; and the metrological tradition codified in the GUM [6] established the principle that measurement uncertainty must be declared before evidence. The frozen-contract discipline inherits from this last tradition directly. The open-source scientific computing ecosystem made the 20,221-test validation suite possible. Reference implementation: https://github.com/calebpruett927/GENERATIVE-COLLAPSE-DYNAMICS.

Appendix A: Heterogeneous Counterexample

To demonstrate why the integrity bound is a diagnostic and not merely an inequality, we contrast the homogeneous trace with two heterogeneous cases.

1. Moderate heterogeneity

Replace $\mathbf{c} = (0.99, 0.99, 0.99)$ with $\mathbf{c} = (0.85, 0.92, 0.98)$—the $t = 0$ state from the end-to-end reference casepack `UMCP-REF-E2E-0001`:

Output	Homogeneous	Heterogeneous
F	0.9900	0.9167
ω	0.0100	0.0833
κ	-0.01005	-0.08870
IC	0.9900	0.9151
Δ	0.0000	0.0015
S	0.0560	0.2665
C	0.0000	0.1062
Regime	Stable	Watch

The heterogeneous trace drops to Watch because $\omega = 0.083 > 0.038$ (drift gate fails). The gap appears: $\Delta = 0.0015$.

2. Geometric slaughter

The mechanism of *geometric slaughter* is demonstrated when a single channel approaches zero. Set $c_1 = 0.001$, $c_2 = c_3 = 0.99$:

$$\begin{aligned} F &= 0.6603 \quad \text{(mean still majority-healthy)}, \\ \mathrm{IC} &= 0.0993 \quad \text{(geometric mean destroyed)}, \\ \Delta &= 0.5610 \quad \text{(massive gap)}. \end{aligned}$$

One channel near ε kills IC because the geometric mean multiplies rather than averages. Fidelity F is the arithmetic mean: it is robust to a single outlier. Integrity IC is the geometric mean: it is sensitive to every channel simultaneously. This is why $\mathrm{IC} \leq F$ is not merely an inequality—it is a **diagnostic**. The gap Δ reveals what the arithmetic mean conceals.

Appendix B: Degenerate-Limit Derivations

1. Duality identity

The GCD duality $F + \omega = 1$ is definitional ($\omega \coloneqq 1 - F$). Classical unitarity $p + q = 1$ for a Bernoulli trial is the

zero-cost special case where $\Gamma(\omega)$ is removed from the seam budget (20).

2. Integrity bound

For $n = 2$ equal-weight channels, $F = (c_1 + c_2)/2$ and $\mathrm{IC} = \sqrt{c_1 \cdot c_2}$. The condition $\mathrm{IC} \leq F$ is precisely $\sqrt{c_1 c_2} \leq (c_1 + c_2)/2$, i.e., the arithmetic–geometric mean inequality for two variables. For general n with weights, the integrity bound is strictly more general: it carries channel semantics (c_i represents a specific physical quantity), the guard band ($c_i \in [\varepsilon, 1 - \varepsilon]$), and composition laws (IC composes geometrically: $\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$; F composes arithmetically: $F_{12} = (F_1 + F_2)/2$).

The solvability interpretation: given (F, IC) with $n = 2$,

$$c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}, \tag{B1}$$

which requires $\mathrm{IC} \leq F$ for real solutions.

3. Bernoulli field entropy

The GCD entropy (11) is the full Bernoulli field entropy. Each channel c_i is a continuous variable in $(0, 1)$. When the collapse field is restricted to $c_i \in \{0, 1\}$ only, the binary entropy $h(c) = -c \log c - (1 - c) \log(1 - c)$ is evaluated only at the endpoints, and the sum reduces to Shannon's formula for a discrete source with outcome probabilities equal to the weights.

4. Log-integrity relation

$\mathrm{IC} = e^{\kappa}$ is definitional: $\kappa = \sum w_i \ln c_i$ defines κ, and $\mathrm{IC} \coloneqq e^{\kappa}$. The classical exponential map $e^x : \mathbb{R} \to \mathbb{R}^+$ is what remains when the weighted sum is replaced by a scalar and the channel structure is removed.

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